A MONOMORPHISM TOWARDS A LOCALLY QUASI-FINITE SPACE OVER A FIELD IS LOCALLY QUASI-FINITE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-over-fields.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the decomposition of the target into Artinian pieces and the reduction to the preimages of those pieces, the analysis of a one-point preimage through the epimorphism of zero dimensional local rings, and the application of Nakayama's lemma to obtain finiteness over the field.

Editorial note. The label lemma-mono-towards-locally-quasi-finite-over-field names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. Pre-open gate: python operations/agent-authoring/tools/precard_gate.py –source spaces-over-fields.tex –label lemma-mono-towards-locally-quasi-finite-over-field –start 2061 –end 2086 –local-id 28064 returned RESULT=OK (L28064 free, occupancy FREE, public interval clear). Candidate taken from sources/stacks-spaces-over-fields/candidates-halfA.json.

Editorial note. The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Structure of a locally quasi-finite space over a field. Same chapter, the lemma describing a locally quasi-finite algebraic space over a field as a disjoint union of spectra of Artinian local algebras finite over the field. Supplies the decomposition of the target into the pieces Spec(A_i).
- Monomorphisms towards a disjoint union of zero dimensional rings are schemes. Decent Spaces, the lemma that the source of a monomorphism towards such a disjoint union is a scheme. Supplies the conclusion that X is a scheme, so that the preimages of the pieces can be treated as affine schemes.
- Nakayama's lemma. Algebra, the lemma allowing the surjectivity of a map of modules to be checked modulo a nilpotent ideal. Supplies the surjectivity of the ring map from the Artinian local ring to the local ring of the preimage from the equality of residue fields.

Source reference. Original lemma 00DV: sources/stacks-chow/tags.txt, source lines 512–512. Prerequisite pointer invoked at proof line(s) 2081; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:spaces-over-fields.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 06RZ: sources/stacks-chow/tags.txt, source lines 8658–8658. Prerequisite pointer invoked at proof line(s) 2073; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 06S0: sources/stacks-spaces-over-fields/pinned-original-source.tex, source lines 2034–2034. Prerequisite pointer invoked at proof line(s) 2069; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let k be a field. Let $f : X \to Y$ be a monomorphism of algebraic spaces over k. If Y is locally quasi-finite over k so is X.*

Proof. Assume Y is locally quasi-finite over k. By Lemma 06S0 we see that $Y = \coprod \mathrm{Spec}(A_i)$ where each A_i is an Artinian local ring finite over k. By Decent Spaces, Lemma 06RZ we see that X is a scheme. Consider $X_i = f^{-1}(\mathrm{Spec}(A_i))$. Then X_i has either one or zero points. If X_i has zero points there is nothing to prove. If X_i has one point, then $X_i = \mathrm{Spec}(B_i)$ with B_i a zero dimensional local ring and $A_i \to B_i$ is an epimorphism of rings. In particular $A_i/\mathfrak{m}_{A_i} = B_i/\mathfrak{m}_{A_i}B_i$ and we see that $A_i \to B_i$ is surjective by Nakayama's lemma, Algebra, Lemma 00DV (because $\mathfrak{m}_{A_i}$ is a nilpotent ideal!). Thus B_i is a finite local k-algebra, and we conclude by Lemma 06S0 that $X \to \mathrm{Spec}(k)$ is locally quasi-finite. □